Table of Contents

Abstract **4**

1.0 Generative Research **5**

1.1 Research Question **5**

1.2 Research Method **7**

1.3 Sample Size **8**

1.4 Participant Details **9**

1.4.1 Explorer 9

1.4.2 Guide 13

1.4.3 Equipment Provider 16

2.0 Affinity Analysis **21**

2.1 First Level Quotes **21**

2.1.1 Explorer 21

2.1.2 Guide 24

2.1.3 Equipment Provider 26

2.2 First Person Summarized Notes **28**

2.2.1 Explorer 28

2.2.2 Guide 30

2.2.3 Equipment Provider 32

2.3 Declarative Statements on User Behaviours **34**

2.3.1 Explorer 34

2.3.2 Guide 34

2.3.3 Equipment Provider 35

2.4 Second Level Notes **36**

2.4.1 Explorer....36
2.4.2 Guide....37
2.4.3 Equipment Provider....39
2.5 Third Level Notes....41
2.5.1 Explorer....41
2.5.2 Guide....41
2.5.3 Equipment Provider....42
2.6 Affinity Chart....43
2.6.1 Explorer....43
2.6.2 Guide....44
2.6.3 Equipment Provider....45
2.7 Design Ideas for each User Role....46
2.7.1 Explorer....46
2.7.2 Guide....47
2.7.3 Equipment Provider....48
3.0 References....49
4.0 Appendix....50
4.1 Semi Structured Interview....50
4.1.1 Interview Consent Request and Context....50
4.1.2 Interview Questions....52
4.1.3 Interview Closing Statement....54
4.1.4 Interview Outcomes....55
4.2 Work Breakdown Structure....69

Abstract

PathFindr is an outdoor adventure planning application designed to connect explorers, certified guides, and equipment providers through a centralized digital platform. The application aims to bridge the gaps present in the existing systems by enhancing the overall experience for explorers when booking tours and guides, assisting guides in managing their clients, and supporting equipment providers in handling their business operations more efficiently.

In this document, the details of the generative research will be outlined, including the research questions, methodology, sample size, and participant details. This is followed by an affinity analysis, which consists of a three-layer data hierarchy and will be outlined in detail. Lastly, the design ideas derived from the collected data will be presented.

1.0 Generative Research

1.1 Research Question

1. What are the most important factors that Explorer look for when planning an outdoor adventure, and what challenges do they face?

This research question aims to gain insights into users' preference and pain points when discovering and planning outdoor activities. It also seeks to identify user's behaviour and habits. By understanding these challenges, it allows the team to design features that simplify the search process and provide a straightforward solution that directly address these pain points. In the meantime, this will help PathFindr position itself as a more user-friendly alternative compared to other software or search engines.

2. What are the key features that users consider most important in an outdoor activity planner?

To ensure that PathFindr meets and exceeds user expectations, it is important to understand which features and functionality users prioritize when using similar platforms. Users might value different features, including activity suggestions, user reviews or weather forecasts. By gathering this information, PathFindr can focus development efforts on the features that user will find most beneficial, helping to avoid investing resources in features that may not significantly impact user satisfaction.

3. What factors influence users' decision to rent or buy equipment through an app instead of offline platforms?

As PathFindr offers the option to buy or rent equipment, it is important to understand what are the factors that might influence user decisions, such as price, convenience, product availability and the ease of delivery or pickup. This information will help PathFindr to design its term and condition to address users's concerns by providing a clear pricing, reliable reviews and flexible delivery option. Besides, it will also ensure PathFindr had provides a clear advantage over traditional purchase method.

4. How can PathFindr integrate community-based features like user review to enhance user experience?

Community and social features can play a significant role as they help build trust and create a sense of belonging among users. Furthermore, outdoor enthusiasts often rely on peer recommendations and reviews to make informed decisions. By integrating user review and community forums, PathFindr can foster a sense of community and encourage repeat use. This can also promote authenticity, encourage more active participation, and improve long-term user retention.

5. What are the key safety concerns users have when planning outdoor activities, and how can PathFindr address them?

The purpose of this question is to identify user's safety-related concerns such as trail difficulty, weather conditions, medical emergencies, or equipment reliability. This question aims to uncover these concerns by suggesting features like real-time alert or SOS notifications. Addressing these concerns not only build user trust but also position PathFindr as a responsible and reliable outdoor activity planner, ultimately enhancing the overall user experience.

1.2 Research Method

Semi-structured interviews are selected as the most appropriate research method for this project due to its adaptability and ability to yield rich, in-depth qualitative data. This technique involves the use of a thematic guide consisting of open-ended questions, while allowing the interviewer to vary the wording, order, and ask follow-up questions based on the participant's responses. This blend of structure and flexibility enables the collection of both consistent and detailed data, making it possible to explore participants' perspectives in their own words (Jamshed, 2014; Kallio et al., 2016).

It is deemed suitable for this study, which involves understanding the experiences of multiple roles, explorer, guide and equipment provider which each has different and distinct concerns. The flexible format enables the interviewer to delve deeper into role-specific insights, clarify responses, and establish rapport, fostering open and honest communication (Gill et al., 2008; Rabionet, 2014). Literature consistently highlights these advantages as key reasons for using semi-structured interviews in diverse fields, including healthcare, education, and user experience research (DiCicco-Bloom & Crabtree, 2006).

Given that the project is in its early, exploratory phase, it is essential to gather contextualized narratives and direct quotations to support affinity mapping and persona development. Semi-structured interviews are ideally suited for such generative research, allowing researchers to build on prior knowledge while remaining open to uncovering unexpected themes, motivations, and language directly from participants. Developing and piloting the interview guide, as recommended by Kallio et al. (2016), ensures methodological rigor and enhances the validity of the data collected.

The method also offers practical benefits that align well with the project's constraints. It supports small, purposive samples and is efficient to conduct, whether in person, online, or by phone, making it ideal for projects with limited time and resources. Studies in applied research contexts have praised semi-structured interviews for their ability to reveal beliefs and behaviors in a time-effective and resource-conscious manner, reinforcing their suitability for this project (Gill et al., 2008; Jamshed, 2014).

1.3 Sample Size

In this study, a total of 9 participants were selected, with 3 individuals representing each of the key roles intended for the proposed PathFindr system: explorer, guide, and equipment provider. Since the target audience of this system is Malaysians who have an interest in outdoor adventures, whether as adventures, guides, or equipment sellers, it is important to select participants who have relevant backgrounds. This approach ensures that they have sufficient understanding and insight into the limitations and challenges of the existing systems, allowing them to provide their feedback for the development of PathFindr.

Moreover, our team will encourage the participants to share their ideas and opinions in open-ended discussions while enabling the researchers to explore their perspectives in greater depth. This ensures that the participants have the opportunity to express their role-specific needs and highlight the potential usability challenges that the PathFindr system should address.

1.4 Participant Details

When selecting the participants, it is crucial to choose the individuals who have relevant knowledge and experience in outdoor adventures. This approach ensures that the data collected is reliable, while also helping to align the proposed system with the specific needs and expectations of its intended users.

1.4.1 Explorer

An explorer is a user role that represents the individuals who are willing or interested in discovering outdoor adventure locations and exploring them. While they may not be experts in outdoor activities, they have a genuine interest in such experiences. Therefore, when selecting the participants for the Explorer role, it is crucial to ensure that they have some experience with outdoor adventures and have previously booked a guide. This approach helps ensure that the feedback they provide is relevant and meaningful, based on real interactions with adventure-related services.

Participant ID	Name	Age	Experience Level	Frequency of Outdoor Activities	Background Info
E01	Nur Aisyah	25	Beginner	Every 2-3 weeks	Enjoys light hiking and has joined beginner nature tours.
E02	Ahmad Zulkarnain	28	Intermediate	Monthly	Loves exploring caves in Malaysia and often uses social media to book guides.
E03	Tan Jia Ming	31	Beginner	Occasionally	Frequently joins comm unity hiking groups with guide support.

Figure 1

Participant Profile - Nur Aisyah



NUR AISYAH

BEGINNER-LEVEL NATURE HIKER

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BACKGROUND

Nur Aisyah is a 25-year-old beginner hiker who enjoys light outdoor activities and nature exploration. She typically goes on hikes every 2-3 weeks. As a beginner in hiking, Aisyah prefers safer, well-marked, and casual outings exploration places. She values experience that helps her connect with nature without requiring advanced physical endurance or technical knowledge.

PROBLEM ADDRESSED

Pain Points

- Difficulty finding suitable trials that match her skill level.
- The lack of safety features in existing applications makes her feel unsafe and unsure if she can rely on them.
- She feels confused when preparing equipment due to her limited knowledge and experience in this area.
- Lack of accessible group hiking options.

Needs & Opportunities

- Need for a system that can recommend beginner-friendly trails tailored to her experience level.
- Require an SOS alert feature that allows her to notify the guide in case of an emergency.
- Want a system that can recommend the necessary equipment based on the selected trail.

SUMMARY

As a beginner in hiking, Aisyah consistently requires more detailed and easy-to-understand guidance. During the interview, she shared that she struggles with existing applications, as many of them overlook the users with low skill levels. She also highlighted several issues, including difficulties in booking reliable guides, preparing appropriate equipment, and feeling safe during hikes. Lastly, Aisyah hopes that the proposed system will address these needs and enhance her overall adventure experience.

Figure 2

Participant Profile – Ahmad Zulkarnain



AHMAD ZULKARNAIN

INTERMEDIATE-LEVEL CAVE EXPLORER

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BACKGROUND

Ahmad Zulkarnain is a 28-year-old intermediate-level cave explorer. He typically explores caves in Malaysia on a monthly basis. As someone who enjoys social group activities, he frequently uses social media to find guides and connect with exploration teams. With his growing experience, Ahmad is now looking for a more organized and trustworthy platform that can help him book guides efficiently while enhancing safety during his adventures.

PROBLEM ADDRESSED

Pain Points

- Lack of a centralized guide booking system forces him to navigate through multiple social media platforms to find a guide.
- Difficult to compare guides and packages as the information on social media is scattered and inconsistent.
- Uncertainly about equipment quality and reliability.

Needs & Opportunities

- Need for a system that centralized guide and package information for easier comparison and booking.
- Require a platform that features only licensed guides to ensure safety and credibility.
- Require a transparent gear review system that allows him to view ratings before renting or purchasing, and leave feedback after use.

SUMMARY

As an individual who enjoys socializing, finding a suitable guide and adventure group is a top priority for him. However, he faces significant challenges when browsing and booking guides, particularly in comparing different packages and services. He also expressed a strong preference for booking only licensed guides to ensure safety and reliability. While he has some experience with preparing gear, he struggles to identify which equipment is high quality and trustworthy.

Figure 3

Participant Profile – Tan Jia Ming



TAN JIA MING

BEGINNER-LEVEL HIKER

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BACKGROUND

Tan Jia Ming is a 31-year-old beginner hiker who participated in hiking activities occasionally. As a beginner. He frequently joins community hiking groups that provide guide support, as he feels more confortable exploring nature in a group. Besides that, he also prefers organized outings where he can rely on experienced guides for safety and direction.

PROBLEM ADDRESSED

Pain Points

- Lack of safety support and awareness features in current applications makes him feel unsafe.
- Difficult to determine the trustworthy guides due to the absence of guide reviews.
- Uncertainly and bad experiences with gear preparation.

Needs & Opportunities

- Require in-app safety tools such as checklists and SOS emergency features.
- Need for a guide review and rating system
- Require a gear rental system that recommends high-quality equipment and includes transparent user ratings.

SUMMARY

As a middle-aged beginner hikes, Jia Ming places a strong focus on safety during his hikes. He has previously experienced purchasing poor-quality or broken gear, leading to now he actively seeks reviews before making any gear-related decisions. In addition, he prefers to book highly rated guides and hopes for a system that offers transparent guide ratings. Jia Ming also wishes the app could provide clear safety guidelines to help him feel more prepared and secure during his hiking experiences.

1.4.2 Guide

A Guide is responsible for leading Explorers during their outdoor adventure experiences. This user group may use their free time to offer guiding services as a way to earn additional income. In addition to having practical experience, they are expected to have expertise in the specific environments they guide others through. By conducting interviews with guides, researchers can gain insights into the challenges they face, such as difficulties in finding customers, issues encountered during exploration, and other role-specific concerns.

Participant ID	Name	Age	Years of Experience	Frequency of Guiding	Background Info
G01	Chong Li Xian	32	5 years	Weekly	Certified outdoor guide specializing in cave exploring.
G02	Mohd Azri	30	7 years	Weekly	A well-known jungle guide in Perak with a spotless safety record.
G03	Hafiz Rahman	40	10 years	Seasonally	Blends outdoor activities with cultural exploration in Penang.

Figure 4

Participant Profile – Chong Li Xian



Figure 5

Participant Profile – Mohd Azri



Figure 6

Participant Profile – Hafiz Rahman



1.4.3 Equipment Provider

When explorers plan their outdoor adventures, they may not always have the necessary equipment. As a result, they need to obtain it from equipment providers. In the PathFindr system, Equipment Providers play the role of offering both sales and rental services to explorers. This not only supports the needs of the explorers but also provides an additional income opportunity for the providers.

Participant ID	Name	Age	Type of Equipment Offered	Platform Used for Business	Background Info
EP01	Goh Mei Ling	37	Water activity gear	Instagram, Facebook	Owns a shop in Kuala Lumpur that specializes in selling high-quality water sports equipment.
EP02	Ali Hakim	32	Climbing gear	Lazada, Shopee Marketspace	Operates in popular climbing destinations.
EP03	Syed Nasrul	29	Camping kits, lanterns, cookware	Carousell, Facebook Marketspace	Experienced in supplying camping kits tailored for beginners and families.

Figure 7

Participant Profile – Goh Mei Ling



GOH MEI LING

WATER SPORTS ENTREPRENEUR

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BACKGROUND

Goh Mei Ling is a seasoned water sports entrepreneur who owns a shop in Kuala Lumpur that specializes in high-quality water activity gear such as paddleboards, life vests, and kayaking equipment. She actively promotes her business through Instagram and Facebook, engaging directly with outdoor enthusiasts and beginner adventurers. Her insight is rooted in both customer interaction and gear performance, especially for water-based activities.

PROBLEM ADDRESSED

Pain Points

- Lack of clear categorization for gear (beginner vs. advanced), which confuses customers.
- No visibility into maintenance status, especially critical for inflatable gear.
- Lack of direct communication (e.g., chat function) to inquire about water or weather conditions.

Needs & Opportunities

- Add tags or filters for beginner-friendly vs. advanced gear.
- Show maintenance and condition status for each item, especially inflatable equipment.
- Integrate a chat function for real-time assistance on gear and environmental conditions.

SUMMARY

Goh Mei Ling has brings valuable insight from both user interactions and gear performance. Her main pain points include the lack of clear gear categorization (beginner vs. advanced), limited visibility into maintenance status—particularly for inflatable gear—and the absence of a direct communication channel for real-time inquiries about weather or water conditions. She sees opportunities in adding tags or filters to better guide users, displaying the maintenance and condition status of gear, and integrating a chat feature to enhance customer support.

Figure 8

Participant Profile – Ali Hakim



ALI HAKIM

CLIMBING GEAR PROVIDER

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BACKGROUND

Ali Hakim is a climbing gear provider who operates around popular climbing destinations in Malaysia. He sells his products through Lazada and Shopee Marketspace, focusing on accessibility and convenience for climbers of all levels. With hands-on knowledge of outdoor climbing needs and gear durability, Ali brings a practical and market-driven perspective to gear planning and rental systems.

PROBLEM ADDRESSED

Pain Points

- Limited ability to showcase gear suitability (e.g., for beginners or experts) in a crowded marketplace.
- May struggle with real-time inventory updates, leading to potential overbooking or customer dissatisfaction.
- Difficult to manage customer feedback and reviews effectively through third-party platforms.

Needs & Opportunities

- Features like gear labeling and categorization (e.g., beginner, intermediate) to help customers choose appropriate items.
- Tools for real-time inventory and order management to prevent overbooking or confusion.
- Integration of user reviews and ratings specific to outdoor activity equipment to build trust and encourage repeat purchases.

SUMMARY

Ali Hakim, a climbing gear provider operating via Lazada and Shopee, prioritizes accessibility and convenience for climbers of all levels. His main challenges include difficulty showcasing gear suitability, managing real-time inventory, and handling customer feedback on third-party platforms. He sees value in features like gear labeling by skill level, real-time inventory tools, and integrated user reviews to improve trust and customer experience.

Figure 9

Participant Profile - Syed Nasrul



2.0 Affinity Analysis

2.1 First Level Quotes

2.1.1 Explorer

E01 – Nur Aisyah	
1	“I love easy hikes, but I often struggle to find the trails that match my skill level”
2	“I wish the app could recommend beginner-friendly trails for me.”
3	“I want a packing list that suites the trail.”
4	“Sometimes I worry if the trail will be crowded and it puts me off going.”
5	“The existing apps don’t have SOS feature, making me feel unsafe if I get lost on a hike.”
6	“I wish the app had rating features so that I can rate my guide to help others choose better.”
7	“I want to join group hikes but don’t know how to find one that fits my pace.”
8	“Renting gear nearby is confusing, I never know what is available or in good condition.”
9	“Sometimes I feel overwhelmed because I have no way of knowing if the guide is reliable before booking.”
10	“I am never sure what equipment I should bring for hikes. It is confusing without clear guidance.”

E02 – Ahmad Zulkarnain	
1	“I mostly book guides through social media, but I am unsure how reliable they are.”
2	“I often need to switch between multiple websites before I can find a suitable guide.”
3	“I have been forced to cancel trips before because I couldn’t confirm guide in time.”

4	"It is hard for me to compare the prices and packages when every guides uses different platforms."
5	"I have to message each guide individually because there is no centralized system to view their availability."
6	"I find it difficult to review and compare guides, which makes it hard to know who is trustworthy."
7	"The app should let me track my cave progress."
8	"I have to message different guides on social media just to ask for basic info, it is really inconvenient and confusing."
9	"I am not sure which equipment is good or reliable. It would be helpful if the app showed equipment ratings."
10	"It would be nice if I could book licensed guides through the app instead of social media."
11	"I hope I can check the weather forecast when I select a date, instead of having to go to another website."

E03 – Tan Jia Ming	
1	"I love joining community hikes but sometimes it is hard to find groups that match my schedule."
2	"I only have little experiences, so I am not familiar with the safety risks. I feel better if the app included security alerts or reminders."
3	"It is hard to know which guides are reliable for group hikers. I want to see transparent ratings from previous hikers."
4	"I joined a hike once where someone got injured and there was confusion about what to do. I do like guides to have an SOS feature for emergencies."
5	"Sometimes, new hikers join without the right gear or knowledge. It would help if the app had a safety checklist for everyone."

6	“Sometimes I show up underprepared because I did not realize certain equipment was required for the trail.”
7	“I brought a broken gear before. It was a bad experience.”
8	“It would be great if the app showed recommended and highly rated equipment rentals near the trailhead.”
9	“I feel more confident if the app showed real reviews, safety records and verified gear options all in one place.”
10	“I once booked a guide who turned out to be really irresponsible, but there was no way for me to leave a rating or review afterward.”

2.1.2 Guide

G01 – Chong Li Xian	
1	“I get bookings from WhatsApp, Facebook, and SMS—it’s a mess to organize.”
2	“Some people cancel last minute, and I’m already at the cave waiting.”
3	“I’ve guided people who clearly weren’t fit for a cave trek.”
4	“I have a checklist of gear, but I keep typing it out for every new person.”
5	“My tours are in remote areas—signal is bad, so I can’t respond fast.”
6	“Once, someone came with sandals instead of proper boots.”
7	“People ask the same questions: ‘How long?’, ‘What to bring?’, ‘Is it safe?’”
8	“I prefer groups of 4–6, but sometimes 9 people show up unannounced.”
9	“Caves aren’t playgrounds. I need a way to pre-screen clients.”
10	“If someone gets scared or lost in the dark, I have to act quickly.”
11	“I usually give a safety talk before we go in, but some still forget.”
12	“Having a fixed tour status like 'On Trail' or 'Preparation' would help everyone stay informed.”
13	“It would save me time if I could create the tour once, then reuse it.”

G02 – Mohd Azri	
1	“I’ve repeated the same safety briefing at least 100 times.”
2	“Sometimes tourists bring kids without asking if it’s kids-friendly.”
3	“I want to limit group size to 6, but people keep inviting others last minute.”
4	“One hiker wandered off the trail and panicked—it took 45 minutes to find him.”
5	“I’m proud of my spotless safety record, but people don’t know about it.”
6	“I use Facebook to post trips, but I still get double bookings.”

7	“I wish there was a live status for the tour like 'On Trail' or 'Completed'.”
8	“Some guests ask for jungle survival tips—I’d love to run short workshops.”
9	“I prefer early morning hikes, but I struggle to communicate time zones.”
10	“I want to build a reputation as a reliable guide, not just a name on Facebook.”
11	“Every trip, I talk about leeches, hydration, trail behavior—same script.”
12	“Group leaders should be able to check who has joined in real time.”
13	“It would help to get a quick GPS ping if someone signals an emergency.”

G03 – Hafiz Rahman	
1	“My storytelling makes the hike unique, but I can’t showcase that in current apps.”
2	“Some people only want to exercise, others are here for the stories—I'd love to label my tours.”
3	“I do this part-time, so I need a way to clearly set my seasonal availability.”
4	“I once hosted a tea session after the hike. Everyone loved it—it broke the ice.”
5	“I want to rate clients too—some were disrespectful to the locals.”
6	“I tell guests about local legends, but there's no way to preview this online.”
7	“Penang has beautiful trails, but it’s more than the path—it’s about the history.”
8	“I bring old photos to show people what this place looked like decades ago.”
9	“Some clients came just because they saw my storytelling video online.”
10	“I don’t have time to chat individually with everyone before a tour.”
11	“Pre-hike meetups would help avoid misunderstandings and latecomers.”
12	“A simple status like ‘Meet-up in Progress’ or ‘Tour Completed’ would keep everyone in sync.”
13	“I’ve thought about doing cultural quiz stops along the hike.”

2.1.3 Equipment Provider

EP01 – Goh Mei Ling	
1	"It helps if water gear listings can show whether they are beginner-friendly or for advanced users."
2	"Photos of the gear in real use like paddleboards on water would attract more attention."
3	"Allowing short-term rentals for half-day water activities would be helpful."
4	"There should be real-time updates on gear availability to avoid double booking."
5	"It's important for customers to know safety requirements before renting water gear."
6	"Letting customers choose delivery to beach locations saves time."
7	"Providing bundle sets for snorkelling or kayaking helps user with decision-making and provides a more convenient and less overwhelming process."
8	"There should be a section for maintenance status, especially for inflatable gear."
9	"It is good to allow early pick-up or drop-off time selection."
10	"Having clear return instructions would avoid gear mishandling."
11	"A chat function to ask questions about water conditions would be great."

EP02 – Ali Hakim	
1	"Real-time stock visibility is important since demand spikes during holidays."
2	"It would be useful to offer gear bundles based on climbing routes."
3	"Showing ratings for each gear item helps new climbers make decisions."
4	"Letting customers choose delivery to specific climbing sites is useful."
5	"It's good to allow rental durations for multi-day climbing trips."
6	"Rental gear should include condition notes, especially for helmets and ropes."

7	“Customers should be encouraged to leave reviews after returning items.”
8	“Having a loyalty program for frequent renters encourages repeat business.”
9	“It’s helpful to set weekend-specific rates due to higher demand.”
10	“Climbing gear should be categorized by season or terrain type.”
11	“The system should alert when gear is returned with visible wear.”

EP03 – Syed Nasrul	
1	“Beginner camping sets should be shown clearly with labels.”
2	“Families appreciate gear bundles with everything included.”
3	“There should be options to select gear based on group size.”
4	“Allow customers to choose delivery to campsites or pickup at nearby hubs.”
5	“Rental listings should show photos of what’s inside the camping kits.”
6	“Having safety and user guides in is useful.”
7	“Letting customers extend rentals directly through the app would be convenient.”
8	“There should be tips or blog links for first-time campers.”
9	“Camping gear return checklist helps reduce missing items.”
10	“It is helpful to show weight or size of gear items for packing purposes.”
11	“There should be a section to highlight family-friendly gear.”

2.2 First Person Summarized Notes

2.2.1 Explorer

E01 – Nur Aisyah

As a beginner in hiking, I find it hard to discover trails that match my skill level. I wish there were an app that could recommend the suitable trails based on my experience. Sometimes I also feel confused about what equipment to bring. It would be helpful if the app could generate a packing list and suggest nearby gear rental services that are reliable. In addition, it is also hard for me to find the group hikes that match my pace. Sometimes I also worry about crowded trails, which make me hesitant to go. The lack of an SOS feature in current apps makes me feel unsafe if I get lost. I also wish that I could rate my guide afterward to help others make better decision.

E02 – Ahmad Zulkarnain

I often rely on social media to find and book guides, but I faced significant challenges. I often have to switch between multiple websites and social platforms just to gather some basic information. This is very inconvenient for me, as I need to message each guide individually, compare prices and packages manually, and still be unsure who is reliable, especially when there is no common review system. Sometimes, I had to cancel trips simply because I could not confirm a suitable guide in time. It would be more helpful if I could compare, evaluate and book guides within a single app. I also hope the app can let me to track the progress of my cave tours, and suggest which equipment is reliable or highly rated for the trip. Regarding the weather, I hope the application can provide the forecast for the selected location, which would be very helpful for planning.

E03 – Tan Jia Ming

I like to join community hikes, but it is hard for me to find the groups that match my schedule. As a beginner, I have little knowledge about safety risks, so I wish the app could include alerts or safety reminders. When looking for guides, it is difficult to know how reliable they are, therefore, I hope that the new app allows me to see the transparent ratings from past hikers. I also had a bad experience with the broken gear I bought once. In addition, most

beginners like me don’t know what equipment is best. It would help a lot if the app offered a checklist and recommended highly rated rentals near the trailhead. I have been on a hike where someone got injured, and none of us knew what to do. An SOS or emergency feature would make me feel much safer. Overall, I would feel more confident if the app had real reviews, safety records, and verified gear all in one place. I also like to rate my experiences to help other hikers.

2.2.2 Guide

G01 – Chong Li Xian

I often find myself overwhelmed trying to manage bookings across several platforms like WhatsApp, Facebook, and SMS. It's not only inefficient, but stressful when clients cancel last-minute or show up unprepared. I regularly resend the same packing list and respond to repetitive questions about duration, safety, and gear. Because my tours are in caves with poor reception, I can't always respond to messages promptly. Sometimes people come unequipped, like wearing sandals instead of hiking boots, or they're physically unfit for the intensity of the cave environment. I prefer smaller groups, but miscommunication often leads to larger ones. A system that helps me manage bookings, screen clients, and keep everyone updated on the tour status would significantly reduce risks and save time.

G02 – Mohd Azri

As a jungle guide, I'm constantly repeating the same safety briefings, which takes up time and energy before every tour. I want tighter control over my group sizes and clearer communication when clients bring additional people, or worse, young children to non-suitable hikes. My record is spotless, but there's no real way to show that to new clients and build their trust. I've had hikers wander off the trail, leading to long delays and safety concerns. I would really benefit from an emergency GPS alert system and a reputation profile to back my credibility. Managing bookings manually results in double entries, and many people forget to leave feedback. I want to build a consistent professional image while reducing the chaos around scheduling, safety, and client expectations.

G03 – Hafiz Rahman

For me, guiding is as much about storytelling and culture as it is about the trail itself. But I struggle to convey this unique value through existing platforms. Many clients don't realize what I offer until the hike begins. I'd love to tag my tours with themes like "history-focused" or "cultural heritage" to better match with the right audience. Since I only guide groups seasonally, clear availability settings would help avoid confusion. I've also hosted casual tea meetups before hikes, which made the experience smoother and more personal. I

don’t have time to chat individually with every client before a tour, so pre-hike events and a shared status update system would help set expectations. Ratings, storytelling modules, and cultural previews would let me grow my brand while keeping logistics manageable.

2.2.3 Equipment Provider

EP01 – Goh Mei Ling

As a shop owner who sells water sports equipment, I believe that offering gear with clear labels indicating whether it is beginner-friendly or suited for advanced users would be really helpful. Besides, adding real-life photos, such as paddleboard being used in the water would also attract more attention from customers. Moreover, a feature for short-term rentals, such as half-day options would make it more flexible. In order to avoid confusion and misunderstanding, real-time updates on gear availability are crucial to avoid double bookings. It's also important to inform customers about safety requirements before they rent gear, and offering delivery options directly to beach locations would save time. Bundles for activities like snorkelling or kayaking would help users, especially beginners, know what gear is required, making the process more convenient and less overwhelming. Besides, a maintenance status section, especially for inflatable gear, would ensure customers are aware of the gear's condition. Offering the ability to select early pick-up or drop-off times and providing clear return instructions would also help prevent gear mishandling. Lastly, I think a chat function would be a great addition for customers to ask questions regarding water conditions.

EP02 – Ali Hakim

I find that real-time stock visibility is essential, especially since demand spikes during holidays. Additionally, I think that offering gear bundles tailored to specific climbing routes would simplify the selection process for climbers. Ratings for each item would also be especially helpful for new climbers when making decisions. A delivery option to specific climbing sites would be convenient, as would the ability to rent gear for multiple days. I think that it is also crucial to include condition notes for rental gear like helmets and ropes to ensuring my customers' safety. In terms of building trust and transparency, it would be encouraging to request customers to leave reviews after returning items. A loyalty program for repeat customers would also definitely encourage business growth, and weekend-specific rental rates are necessary to reflect higher demand. I also think it would be useful to categorize climbing gear by season or terrain type. Finally, a system alert for gear returned with visible wear would help maintain the quality of the rental stock.

EP03 – Syed Nasrul

When it comes to camping gear, clearly labelling beginner-friendly sets would make it easier for new campers to find what they need. Besides, there should also be options to select gear based on group size, as this helps to tailor the rental to the customer's needs. In terms of delivery services, a delivery option directly to campsites or the ability to pick up gear at nearby hubs would provide convenience. Rental listings should also include photos of what's inside the camping kits, so customers know exactly what they're getting. It's also important to provide safety and user guides with the gear, particularly for first-time campers. Offering an option to extend rentals directly through the app would be a practical feature. Meanwhile, I believe having a checklist for gear returns would reduce the chances of missing items. Additionally, showing the weight or size of gear items would be helpful for customers packing for their trips, and highlighting family-friendly gear could attract more customers with children.

2.3 Declarative Statements on User Behaviours

2.3.1 Explorer

Based on insights from the three explorers, it was revealed that users commonly face challenges in discovering trails and group hikes that match their skill levels, which may cause hesitation or lead them to cancel trips. Currently, most explorers rely on social media to find and book guides, often resulting in confusion, delayed decisions, and difficulty assessing guide reliability due to the lack of a unified review system. They also wish for the application to provide weather forecasts directly, rather than requiring them to visit external websites to obtain this information. Regarding equipment, beginners often need recommendations, as they lack knowledge about appropriate gear and safety practices. They responded that it would be helpful if the app provided a built-in curated packing checklist, verified rental options, and transparent equipment ratings. Beginners also frequently express concerns about safety on trails and hope the system can include features such as real-time tracking, SOS alerts, and accessible safety information. In conclusion, the explorer user group hopes to have an application that simplifies trip planning, builds trust in guides and gear, and enhances their sense of safety and preparedness.

2.3.2 Guide

Guides currently struggle with fragmented communication channels, often juggling multiple apps and repeated manual tasks just to coordinate a single tour. Many find themselves answering the same questions repeatedly, managing cancellations, and sending redundant checklists or safety tips. There is a clear desire for automation and centralized control over booking logistics. Furthermore, guides express strong concern over mismatched client expectations—be it fitness levels, group size, or preparedness—and would benefit from pre-screening tools or templated briefings. Beyond logistics, guides want to showcase what makes their services unique, whether it’s a flawless safety record, immersive storytelling, or local cultural knowledge. Seasonal guides need flexible availability settings, and all guides welcome tools for live trip status tracking and emergency alerts. A clear pattern emerges where guides want to spend less time managing logistics and more time delivering enriching, safe, and personalized outdoor experiences.

2.3.3 Equipment Provider

All the business owners and equipment providers emphasize the importance of a clear and informative labels, such as “beginner-friendly” or “advanced” to assist customer in choosing the right gear. Additionally, they find that providing real-life images of the equipment makes the selection more appealing. Real-time stock visibility and updates are also critical, especially during peak demand periods, to prevent double bookings and ensure the customers have access to the equipment they need. On the delivery side, offering different delivery options to specific locations adds significant convenience and greatly enhance the customers experience and satisfactions. Apart from that, providing flexible rental options such as short-term or multiple day rentals also adds value by catering different user needs. Moreover, the importance of safety and condition transparency is also emphasized, with suggestions as including out details such as condition notes and safety guides when selling or renting the equipment. Providing bundles tailored to specific activities, along with loyalty programs as well as customer reviews also helps in build trust and encourage repeat business. In further enhance user experience, it is also important to allow customer to customize their rental terms, including early pick-up and extend rentals. Finally, having a checklist for returns and alerts for worn equipment could help in maintaining the quality and reliability of the rental stock.

2.4 Second Level Notes

2.4.1 Explorer

ESL	Second-Level Note	Supporting Quotes
1	Explorers want beginner-friendly trail recommendations that match their skill level, preferences, and hiking pace.	E01:1, E01:2, E01:4, E01:7, E03:1
2	When booking a guide, the explorers wish to review transparent ratings and detailed review, while ensuring the guide holds a valid license.	E01:9, E02:1, E02:6, E02:10, E03:3
3	Explorers wish to access the weather conditions directly through the application.	E02:11
4	Explorers often face difficulties in comparing guides and booking trips due to scattered information.	E01:9, E02:2, E02:3, E02:4, E02:5, E02:8
5	Explorers face difficulties in preparing their gear and wish to have personalized packing lists to guide them.	E01:3, E01:10, E03:6
6	Explorers wish the app to provide clear and trusted gear rental options, featuring high-quality equipment.	E01:8, E03:7, E03:8
7	Explorers want to track their trip progress and access tools that enhance their adventure experience.	E02:7, E03:9
8	Explorers wish the app to enhance their safety by providing features such as SOS alerts, emergency information, and safety checklists.	E01:5, E03:2, E03:4, E03:5
9	Explorers wish to review the ratings of guides and gear before making a purchase, and to leave their own ratings afterwards.	E01:6, E2:9, E3:10

2.4.2 Guide

GSL	Second-Level Note	Supporting Quotes
1	Difficulty managing and organizing bookings across multiple platforms.	G01:1, G02:6, G03:10
2	Last-minute changes, no-shows, or unannounced guests cause logistical problems.	G01:2, G01:8, G02:2, G02:3
3	Current systems lack proper safety and emergency tools.	G01:10, G02:4, G02:13
4	Inadequate screening of participants leads to safety and compatibility issues.	G01:3, G01:6, G01:9, G02:2
5	Guides frequently repeat the same information, like safety briefings or FAQs.	G01:4, G01:7, G01:11, G02:1, G02:11
6	Need for a tour status system to keep everyone informed in real-time.	G01:12, G02:7, G03:12
7	Lack of automated or reusable tour setup increases manual workload.	G01:4, G01:13, G03:3
8	Communication challenges are due to remote locations, poor signal quality, and time zones.	G01:5, G02:9
9	Guides desire a system to pre-screen or qualify participants.	G01:9, G03:5
10	Interest in offering personalized, thematic, or educational experiences.	G03:1, G03:2, G03:6, G03:7, G03:13
11	Need for guide branding, client feedback, and reputation-building tools.	G02:5, G02:10, G03:5, G03:9
12	Interest in social/community features like pre-hike meetups or post-tour sessions.	G03:4, G03:11

13	Real-time group management tools would enhance control and coordination.	G02:12, G01:8

2.4.3 Equipment Provider

EPSL	Second-Level Note	Supporting Quotes
1	Equipment providers highlight the importance of having clear labels such as beginner-friendly or advanced to help customers in decision making.	EP01:1, EP03:1, EP03:11
2	Equipment providers wish to have real-life photos of the equipment and details kit contents to improve customer confidence and engagement.	EP01:2, EP03:5
3	Real-time stock visibility and equipment availability updates are essential to prevent overbooking, especially during peak periods.	EP01:4, EP02:1
4	Equipment providers wish to offer geal bundles that are personalized to specific activities like snorkeling and hiking to help users, especially beginners to identify necessary equipment, making the process straightforward and comfortable.	EP01:7, EP02:2, EP03:2
5	Safety instructions, user guide and return checklists improve gear handling and reduce loss or damage.	EP01:5, EP03:6, EP03:9
6	Equipment providers hope to provide flexibility for customers to choose delivery to specific locations such as beaches or campsites for added convenience.	EP01:6. EP02:4, EP03:4
7	Flexible rental options like half-day or multiple-day rental caters to meet and diverse different customer needs.	EP01:3. EP02:5
8	Provide remark like condition notes and maintenance status, especially for critical or inflatable gear, ensuring transparency and safety.	EP01:8, EP02:6

9	Reviews and ratings are significantly importance to help new customers make informed decisions and build trust.	EP02:3. EP02:7
10	Equipment providers wish they could customize the rental terms, including early pick-up or rental extensions, to increase customer satisfaction.	EP01:9. EP03:7

2.5 Third Level Notes

2.5.1 Explorer

ETL	Third-Level Note	Supporting Quotes
1	Streamline trail planning by integrating trail recommendations, weather forecasts, and gear preparation tools.	ESL1, ESL3, ESL5
2	Enhance the guide booking experiences by centralizing listings, showcasing verified credentials, displaying transparent reviews, and integrating availability tools.	ESL2, ESL4, ESL9
3	Build an equipment module that helps explorers with packing and provides access to high-quality, reliable rental options.	ESL5, ESL6
4	Enhance user safety by building features, including SOS alerts and emergency resources.	ESL8
5	Enhance the user experience during adventures by providing real-time tracking of trip progress.	ESL7

2.5.2 Guide

GTL	Third-Level Note	Supporting Quotes
1	Streamline guide operations by consolidating booking, communication, and tour management tools.	GSL1, GSL7, GSL8
2	Build safety-first features, including pre-screening, emergency alerts, and standardized safety briefings.	GSL3, GSL4, GSL5, GSL9
3	Enable dynamic, real-time tour status and group coordination features.	GSL6, GSL13

4	Empower guides to brand their offerings with storytelling, themes, ratings, and content previews.	GSL10, GSL11
5	Support community-driven and social experiences that build rapport and reduce friction.	GSL2, GSL1

2.5.3 Equipment Provider

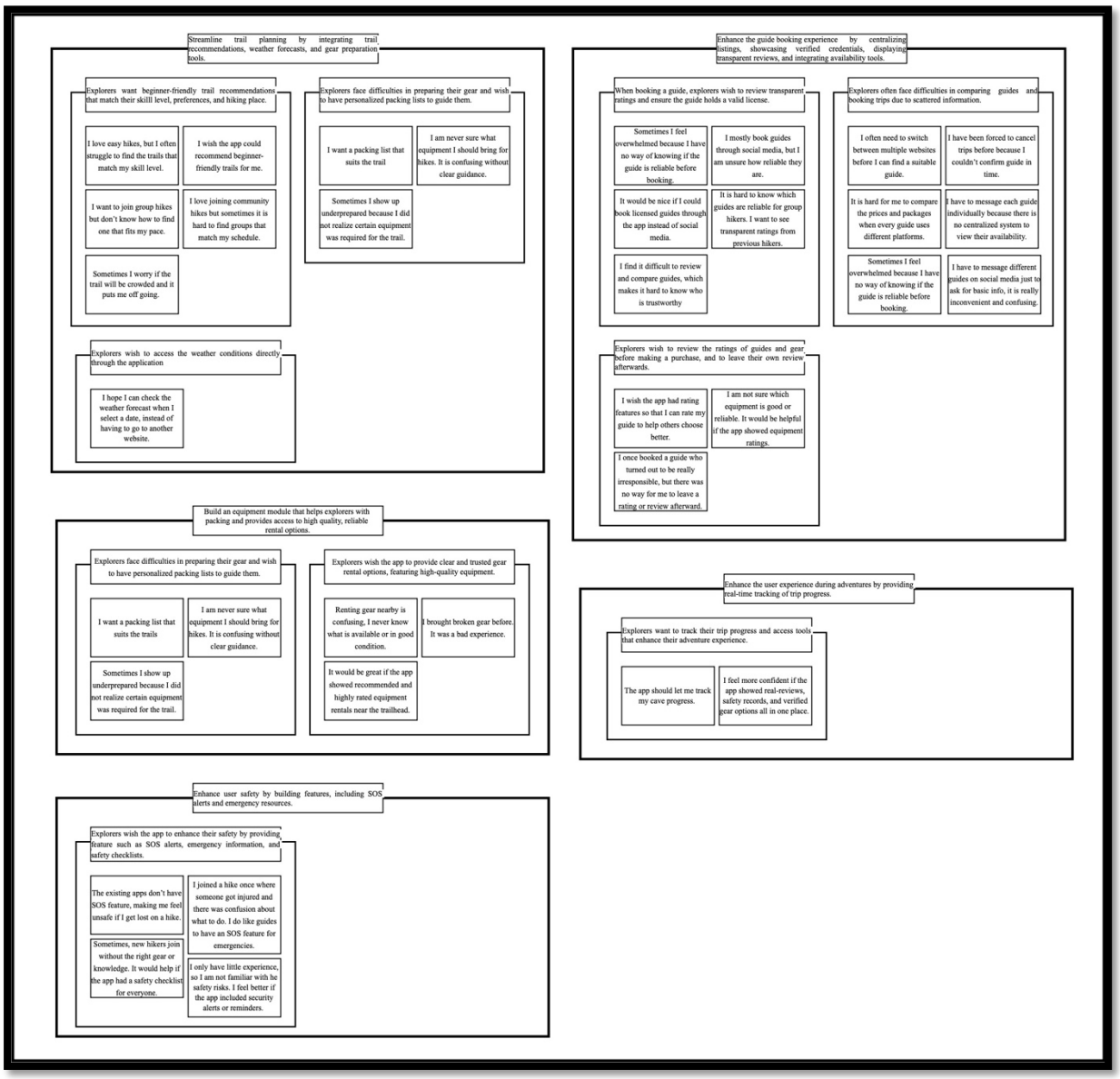
EPTL	Third-Level Note	Supporting Quotes
1	Enable personalized and customized equipment selection by implementing clear labeling, gear ratings and review, as well as details content previews.	EPTL1, EPTL2, EPTL9
2	Improve inventory and availability management by displaying real-time geal updates and condition transparency.	EPTL3, EPTL8
3	Offers personalized rental experience with activity-based bundles and flexible rental durations	EPTL4, EPTL7
4	Enhance convenience through flexible logistics and delivery options.	EPTL6, EPTL10
5	Improve customer safety and equipment accountability through structured guidance, safety instructions and checklists.	EPTL5

2.6 Affinity Chart

2.6.1 Explorer

Figure 10

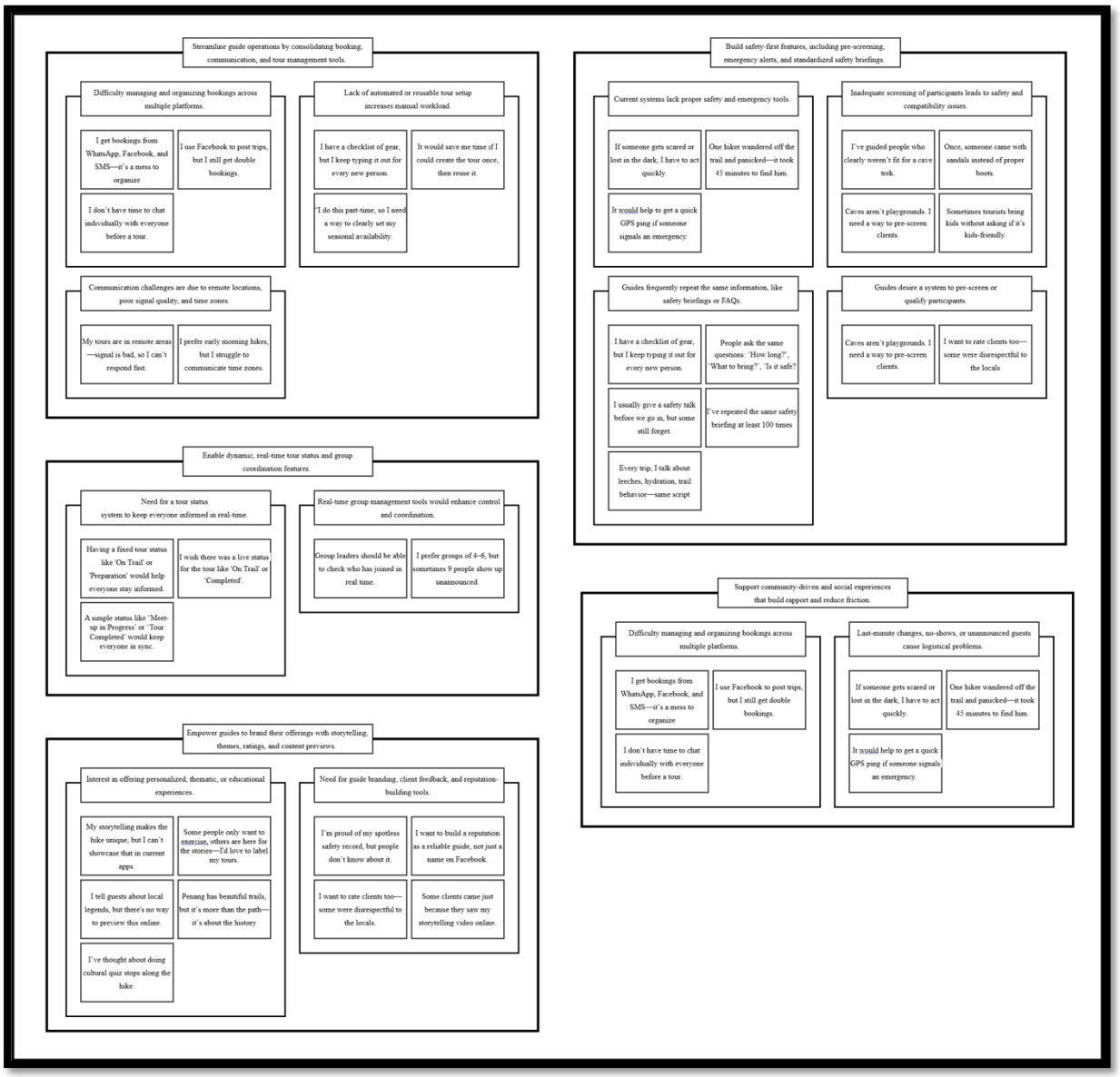
Affinity Chart - Explorer



2.6.2 Guide

Figure 11

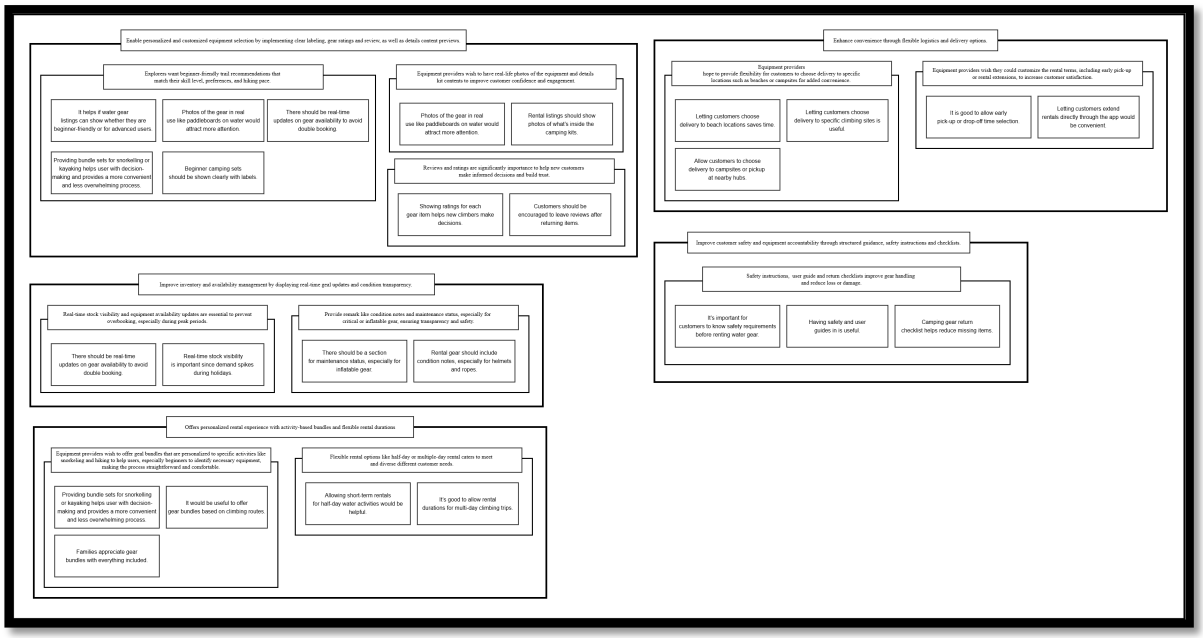
Affinity Chart - Guide



2.6.3 Equipment Provider

Figure 12

Affinity Chart - Equipment Provider



2.7 Design Ideas for each User Role

2.7.1 Explorer

Section	Design Idea
Smart Trail Finder	- Recommend trails based on the skill level and preferences of explorer.
Guide Booking Center	- Allow explorers to browse and compare licensed guides. - Enable explorers to book guides directly through the application. - Allow explorers to view guide ratings and feedback before making a booking.
Weather Monitor	- Provides weather information relevant to the selected location and date.
Gear Assistant Center	- Provide personalized packing checklists tailored to trail. - Enable the user to browse, rent and purchase the equipment from trusted equipment providers. - Display ratings and reviews of rental equipment and providers.
Safety Center	- Introduce an SOS alert feature for emergency situation. - Provide access to emergency contacts information. - Offer a safety checklist to help explorers stay prepared during the trip planning stage.
Trip Tracker	- Enables real-time tracking of trip progress and in-journey updates.
Review & Feedback Hub	- Allow explorers to rate guides and gear after completing a trip or using the equipment. - Enable the explorers to view the community feedback before booking or renting.

2.7.2 Guide

Section	Design Idea
Guide Operations Hub	- Combine and centralize key processes, including tour booking, customer communication, and schedule management.
Guide Safety Toolkit	- Provide pre-tour safety checklists to ensure explorers are well-prepared before starting their journey. - Enable mutual sharing of emergency contact details between guides and participants. - Support standardized safety briefing templates.
Live Tour Dashboard	- Allow the guides to track live tour progress. - Introduce tour phase update features that enable guides to keep participants informed with the latest trip status. - Support group coordination feature to help the guides ensure the safety of participants.
Guide Branding Studio	- Allow the guides to customize their profiles with storytelling elements and thematic branding. - Display guide ratings and media previews to build trust with explorers.
Community Engagement Tools	- Provide in-app messaging features that enable guides to communicate directly with explorers.

2.7.3 Equipment Provider

Section	Design Idea
Smart Equipment Listing	- Provide a feature that allows equipment providers to label their gear. - Allow equipment providers to edit detailed item previews, including descriptions and real-life images. - Introduce an equipment ratings feature to help new explorers make informed decisions.
Real-Time Inventory Dashboard	- Display real-time inventory status, including gear availability to prevent overbooking, - Enable the equipment providers to manage the equipment status, including maintenance updates and condition notes.
Gear Bundles	- Introduce a feature that enables the equipment providers to bundle multiple gear items into activity-based sets.
Flexible Logistics and Delivery	- Enable the equipment provider to offer item delivery service to specific locations. - Provide customization rental terms feature, including early pickup and other flexible rental options.
Safety and Accountability Toolkit	- Introduce a feature that enables equipment providers to upload and manage safety instructions and gear usage guides. - Provide a feature that allows equipment providers to create and share return checklists to explorers.

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4.0 Appendix

4.1 Semi Structured Interview

4.1.1 Interview Consent Request and Context

Before the interview begins, it is crucial to inform the candidates that the session will be recorded and to obtain their consent. This ensures that their privacy is respected and ethical standards are upheld.

To facilitate the process of obtaining consent, a script has been developed:

"Good morning [NAME], thank you for joining the interview session. I am [INTERVIEWER NAME], and I would like to express my sincere gratitude for helping us gather user perspectives. Let begins with introducing the background of our system briefly. Recently, we have revealed that many Malaysians are becoming increasingly interested in outdoor adventure activities to relieve stress. However, many of them face significant challenges when booking guides. For guides, we found that they often struggle to organize and manage their candidates and trails. Similarly, equipment providers have also reported that they face significant challenges in managing their inventory and gaining meaningful insights into their revenue and business performance.

Since you have prior knowledge and experience as a/an [REPRESENTED ROLE], we hope to gather your valuable insight. Before we begin, we would like to request your permission to record the audio of this session for documentation and analysis purposes."

The consent agreements of each candidate are outlined below:

ID	Name	Represented Role	Consent for Recording
E01	Nur Aisyah	Explorer	Recording allowed, no document inclusion
E02	Ahmad Zulkarnain	Explorer	Recording allowed, no document inclusion

E03	Tan Jia Ming	Explorer	Granted
G01	Chong Li Xian	Guide	Recording allowed, no document inclusion
G02	Mohd Azri	Guide	Granted
G03	Hafiz Rahman	Guide	Recording allowed, no document inclusion
EP01	Goh Mei Ling	Equipment Provider	Granted
EP02	Ali Hakim	Equipment Provider	Recording allowed, no document inclusion
EP03	Syed Nasrul	Equipment Provider	Recording allowed, no document inclusion

4.1.2 Interview Questions

4.1.1.1 Explorer

1	How do you find trails that match your skills level or preferences?
2	How do you feel about the way you currently book guides?
3	What has your experience been like when finding and comparing guides online?
4	Do you review ratings or reviews before booking?
5	Do you face any challenges when preparing hear that suits the trail?
6	Have you encountered any issues when purchasing or renting gear?
7	Do you think reviews are a useful feature when renting or purchasing gear?
8	What safety info do you wish apps provided?
9	In your opinion, how can an app help you feel safe during a trip?
10	What feature would your ideal hiking app have?

4.1.1.2 Guide

1	What’s your usual safety routine before hiking?
2	Have you experienced any problems with people not showing up or being late?
3	What challenges have you faced with unprepared or unsuitable clients?
4	Do you use any checklists or guides to help clients prepare?
5	How do you deal with group size and people bringing guests unannounced?
6	Have you had any situations where clients got lost or panicked?
7	How do you currently promote your tours and manage bookings?
8	What tools or features would help you manage a live tour better?
9	What makes your tours unique compared to others in your area?
10	How do you manage your availability, especially if you only guide seasonally?

4.1.1.3 Equipment Provider

1	What kind of information do you think is essential to include in a gear listing?
2	How important do you think photos are when showcasing rental gear? What kind of images do customers expect?
3	Do you think showing the internal contents of gear kits or maintenance status would impact rentals? Why or why not?
4	Do you find it helpful when customers report or record the condition of gear when returning it?
5	What features would you like to see in a rental platform to make your job easier?
6	Do you think bundle gear sets are helpful for users? If so, how should they be grouped or categorized?
7	Do you think a flexible delivery method such as pickup or delivery would help improve your service?
8	What role do you think safety information, return instructions, or checklists play in the rental process?
9	Would it be helpful to have a real-time gear availability system?
10	Would you consider offering features like rental extension through an app, loyalty programs, or weekend-specific pricing?

4.1.3 Interview Closing Statement

After gathering insights from the interviewees, it is crucial and essential to express our gratitude for their time and contribution in sharing their ideas and insights. The script for the interview closing statement is shown below:

"Thank you so much for sharing your valuable insights. This will contribute significantly to our understanding of user needs and the challenges faced. Once again, our team would like to express our sincere appreciation for your time and participation in this interview."

4.1.4 Interview Outcomes

4.1.4.1 Explorer

How do you find trails that match your skills level or preferences?

Nur Aisyah	I love easy hikes, but I often struggle to find the trails that match my skill level. As a beginner, I often look for easy trails, but most of the platforms I use do not clearly label which ones are beginner friendly. Sometimes, I end up on routes that are a bit too challenging.
Ahmad Zulkarnain	For me, I actually like to challenge myself, so I don't mind taking on harder or more challenging routes. However, I face significant challenges when finding the right trail. Currently, I rely on different social media platforms to look for the suitable trails or guides. This is very inconvenient for me because after I find a place, I still have to go to another website just to check the weather. I hope I can check the weather forecast when I select a date, instead of having to go to another website.
Tan Jia Ming	I love joining community hikes but sometimes it is hard to find groups that match my schedule. When I browse trails on social media, there is usually no option for me to filter by time and date, which makes it hard to find something that works for me.

How do you feel about the way you currently book guides?

Nur Aisyah	Sometimes I feel overwhelmed because I have no way of knowing if the guide is reliable before booking. The application I am currently using only shows a basic profile of guides, without any ratings or feedback from past hikers.
Ahmad Zulkarnain	I mostly book guides through social media, but I am unsure how reliable they are. I had past experiences where I encountered scammer, so now I am much more cautions when booking for guide.

Tan Jia Ming	The application I am using currently does not have a review or feedback features, so whether the guide I book is good or not really just depend on luck. I once booked a guide who turned out to be really irresponsible, but there was no way for me to leave a rating or review afterward.

What has your experience been like when finding and comparing guides online?

Nur Aisyah	When I am looking for a guide, I usually will check for the reviews and feedback from previous hikers first before making a decision. However, it is quite hard for me as the app I am currently using does not have a proper review feature. I wish the app had rating features so that I can rate my guide to help others choose better. Whereas for comparing the guides, I do not really have that need.
Ahmad Zulkarnain	It is hard for me to compare the prices and packages when every guide uses different platforms. I really hope there is an app that can centralize all the information and let me compare everything in one place, instead of having to jump between different platforms.
Tan Jia Ming	When it comes to finding a guide, it is hard to know which guides are reliable for group hikes. I want to see transparent ratings from previous hikers. But when it comes to comparing guides, that is not really something I need.

Do you review ratings or reviews before booking?

Nur Aisyah	Participant has already addressed this in Question 3 and would like to skip this question.
Ahmad Zulkarnain	Yes, I do. However, I find it difficult to review and compare guides, which makes it hard to know who is trustworthy. I also not sure which equipment is good or reliable. I believe this issue could be resolved if the app included a review system.

Tan Jia Ming	As I mentioned in the previous question, I wish to know which guide is reliable before booking a guide. I feel more confident if the app showed real reviews, safety records, and verified gear options all in one place.

Do you face any challenges when preparing gear that suits the trail?

Nur Aisyah	As a beginner, I have limited knowledge about the gear. I am never sure what equipment I should bring for hikes. It is confusing without clear guidance. I want a packing list that suits the trail.
Ahmad Zulkarnain	Even though I have prior knowledge what gear I should bring on the trail, but I still have a few concerns. I am not sure which equipment is good or reliable. It would be helpful if the app showed equipment ratings.
Tan Jia Ming	The platform I use to book trails does not show what gear I should bring for the hike. Sometimes I show up underprepared because I did not realize certain equipment was required for the trail.

Have you encountered any issues when purchasing or renting gear?

Nur Aisyah	Renting gear nearby is confusing, I never know what is available or in good condition. I once rented a piece of broken gear, and the shop asked me to pay for the damage. It made me feel really frustrated.
Ahmad Zulkarnain	I am not sure which equipment is good or reliable. It would be helpful if the app showed the equipment ratings. I can rely on the past users' reviews before making decision on purchasing the equipment.
Tan Jia Ming	I bought a broken gear before. It was a bad experience. I ended up canceling my trail because I only realized the gear was broken when I got there.

Do you think reviews are a useful feature when renting or purchasing gear?

Nur Aisyah	As I mentioned just now, reading reviews helps me to understand which equipment is best before renting. I believe this is a feature that all beginner hikers would really appreciate.
Ahmad Zulkarnain	Yes, I think this is a must-have feature in an outdoor adventure planning app. It would be helpful if the app showed equipment ratings. As someone who relies on reviews when renting gear, I would really love that.
Tan Jia Ming	It would be great if the app showed recommended and highly rated equipment rentals near the trailhead. This could help prevent issues like renting broken gear and being forced to cancel a trail.

What safety info do you wish apps provided?

Nur Aisyah	The existing apps don't have SOS feature, making me feel unsafe if I get lost on a hike. It would be really helpful if the app provided this feature.
Ahmad Zulkarnain	I personally consider weather conditions as the primary factor affecting my safety during a trip. I hope I can check the weather forecast when I select a date, instead of having to go to another website.
Tan Jia Ming	I only have little experiences, so I am not familiar with the safety risks. I feel better if the app included security alerts or reminders. It would be really helpful if the app included a safety checklist.

In your opinion, how can an app help you feel safe during a trip?

Nur Aisyah	Sometimes I worry if the trail will be crowded and it puts me off going. It would be helpful if the app could show trail details, including the number of people on the trail.

Ahmad Zulkarnain	I had experiences booking unlicensed guides through social media, and they were very unprofessional during the trip. They often brought us to risky and unsafe place. It would be nice if I could book licensed guides through the app instead of social media.
Tan Jia Ming	I joined a hike once where someone got injured and there was confusion about what to do. I do like guides to have an SOS feature for emergencies. So, in case of an emergency, it is important that we can easily contact the guides to get help.

What feature would your ideal hiking app have?

Nur Aisyah	Sometimes, I want to join group hikes but don't know to find one that fits my pace. Because of that, I had to cancel my plans a few times when I could not find a suitable group. I wish the app could recommend beginner-friendly trails for me.
Ahmad Zulkarnain	I had experiences booking unlicensed guides through social media, and they were very unprofessional during the trip. They often brought us to risky and unsafe place. It would be nice if I could book licensed guides through the app instead of social media.
Tan Jia Ming	Sometimes, new hikers join without the right gear or knowledge. It would help if the app had a safety checklist for everyone. I believe this idea can help ensure everyone understands the safety guidelines and precautions before going on the trip, which can help prevent bad situations from happening.

4.1.4.2 Guide

What's your usual safety routine before hiking?

Chong Li Xian	In my opinion, participants usually only remember the safety precautions right before the trip starts. As a result, I usually give a safety talk before we go in, but some still forget. Unfortunately, this had led to a few of them being ending up in dangerous situation.
Mohd Azri	For every trip, I talk about leeches, hydration, trail behavior – same script. I have repeated the same briefing at least 100 times. I believe this could help prevent them from getting injured.
Hafiz Rahman	I usually go over all the safety precautions during the pre-hike meetups. Pre-hike meetups would help avoid misunderstandings and latecomers. It is a good opportunity to make sure everyone is on the same page before the hike starts.

Have you experienced any problems with people not showing up or being late?

Chong Li Xian	Yes, this happened to me last year. Some people cancel last minute, and I am already at the cave waiting. It was really frustrating because they didn't inform me earlier, and it ended up wasting a lot of my time.
Mohd Azri	Yes, I have experienced issues with people not showing up or being late. I prefer early morning hikes, but I struggle to communicate time zones. This becomes particularly important when planning with participants from different regions, as miscommunication around time zones has led to confusion and delays before.
Hafiz Rahman	Before I started giving safety briefing during the pre-hike meetups, I used to share the safety precautions on the spot. But when latecomers arrived, I had to repeat everything again, which often caused delays in starting the hike. This sometimes made other participants feel frustrated or unsatisfied.

What challenges have you faced with unprepared or unsuitable clients?

Chong Li Xian I had some bad experiences with unprepared and unsuitable clients. I have guided people who clearly were not fit for a cave trek. I had to wait for them one by one before we could move, which made other participants impatient and annoyed. I also have experiences dealing with unprepared clients. Once, someone came with sandals instead of proper bots. I had to cancel the trail because it was too dangerous and could have led to serious incidents. Unfortunately, the participants got angry and even scolded me for the cancellation.

Mohd Azri Sometimes tourists bring kids without asking if it is kid friendly. It is really inconvenient because I have to put in extra effort to ensure the safety of child, which can take attention away from the rest of group.

Hafiz Rahman Sometimes, clients speak inappropriately to the locals, which I find very rude and disrespectful. I want to rate clients too – some were disrespectful to the locals.

Do you use checklists or guides to help clients prepare?

Chong Li Xian Yes, I have used checklists before to help clients prepare the right gear, and I find them very convenient. However, there is no existing app that allows me to share or reuse the checklist easily. I have a checklist of gear, but I keep typing out for every new person.

Mohd Azri No, I haven't really thought about it in that way before. I usually just mention it in my advertisements, but some people tend to ignore it anyway.

Hafiz Rahman No, but that actually sounds like a good idea. Right now, I prepare a PDF document that outlines all necessary gear, but it is quite inconvenient. I don't have time to chat individually with everyone before a tour.

How do you deal with group size and people bringing guests unannounced?

Chong Li Xian I prefer groups of 4-6, but sometimes 9 people show up unannounced. This makes the hike harder to manage and can affect the overall safety and experience for everyone.

Mohd Azri I want to limit group size to 6, but people keep inviting others last minutes. This disrupts my original plan, and I have to quickly come up with a new arrangement on the spot just to make sure everyone can still join the trip.

Hafiz Rahman I will definitely not allow the unannounced guests to join the trip. It is not just about revenue – it is about safety. I know my limit, and I would not take extra risks just to earn more. I have a responsibility to the other participants as well.

Have you had any situations where clients got lost or panicked?

Chong Li Xian Yes, I had an incident last year. One hiker wandered off the trail and panicked – it took 45 minutes to find them. It would be really helpful if there were an app that allowed me to track each client's location during the hike.

Mohd Azri No, I usually count the headcount before moving on to the next stage of the hike. Making sure everyone present is my top priority.

Hafiz Rahman Just last week, one of the participants got lost during the hike. We searched for him for several hours but could not find him, so we eventually had to call for emergency help. Thankfully, he was found alive.

How do you currently promote your tours and manage bookings?

Chong Li Xian Normally, I receive bookings via WhatsApp, Facebook and text message. However, I have found that this makes it difficult to stay organized and track all customer inquiries in one place.

Mohd Azri I used Facebook the most to post about trips, but managing bookings on Facebook was difficult as I encountered issues with double bookings.

Hafiz Rahman	I use storytelling videos to promote my tour. However, most platforms do not allow me to emphasize the unique storytelling aspect of my hikes, which limits my ability to stand out and connect with my audience.

What tools or features would help you manage a live tour better?

Chong Li Xian	In my opinion, having a fixed tour status, such as “On Trail” or “Preparation”, would be helpful as it would keep everyone informed. Other than that, it would reduce confusion and improve communication.
Mohd Azri	It would be helpful for me to have a GPS location, as it would also add a crucial layer of safety support. Besides, having features that shows the real-time status of the tour, such as “Completed” or “On Trail” would also be a bonus.
Hafiz Rahman	Personally, I think a simple live status update, such as “Tour Completed” or “Meet-up in Progress”, would save time and reduce the need for constant manual updates during the trip.

What makes your tours unique compared to others in your area?

Chong Li Xian	I believe that what sets my tours apart is the personal attention I give to each group, from delivering safety talks to carrying out rescues when necessary. I also take extra steps to keep things safe and organised especially when guiding people who are unfamiliar with the area.
Mohd Azri	What makes my tour unique is that I offer hands-on workshops as part of the trip. I also take the safety of my passengers seriously and have a spotless record.
Hafiz Rahman	As mentioned, I would tell stories to share old photographs and social experiences, which some participants would enjoy. Unfortunately, there is no platform that allows me to clearly preview these for the participants.

How do you manage your availability, especially if you only guide seasonally?

Chong Li Xian	In my opinion, it would be much more efficient if I could create a reusable tour listing to remove repetitive setup and make information readily available.
Mohd Azri	The participant has already answer this in Question 7 and would like to skip this question.
Hafiz Rahman	My availability changes with the seasons, but the current system is not user-friendly. I need an easy way to show when I'm available, so that people don't reach out to me during my off-season.

4.1.4.3 Equipment Provider

What kind of information do you think is essential to include in a gear listing?

Goh Mei Ling	I think it is important for gear listing to clearly state whether the equipment is beginner-friendly or mainly for advanced users. Thus, it helps users who are less experienced to make the right choice.
Ali Hakim	For my point of view, it is crucial to include detailed condition notes for each equipment item. I also think categorizing the gear by terrain can help customers to select the right tools for their specific adventure.
Syed Nasrul	For me, I think that clarity is the key as I ensure that the gear listings show whether the gear is meant for beginners. I also prefer to include information and content for each equipment.

How important do you think photos are when showcasing rental gear? What kind of images do customers expect?

Goh Mei Ling	Photos are incredibly important. Based on my experience, images help for customers to visualize the experience and feel confident that what they are buying or renting is functional.
Ali Hakim	The participant has already answer this in Question 1 and would like to skip this question.
Syed Nasrul	I strongly believe that photos should accurately represent the gear kit. That's why I always include a clear image of the items, so the customer knows what to expect. It also helps build trust and prevent confusion.

Do you think showing the internal contents of gear kits or maintenance status would impact rentals? Why or why not?

Goh Mei Ling	Yes, definitely. For me, knowing the maintenance status, especially for inflatable gear, is very important. It gives renters peace of mind that the gear is safe and in good condition.

Ali Hakim	Yes. I believe it is essential as I always want to know the condition of the gear before buying or renting.
Syed Nasrul	I think that it makes a huge difference. When I rent out camping kits, I would include a photo and a checklist to ensure nothing is missing. Including maintenance information also makes the rental process smoother.

Do you find it helpful when customers report or record the condition of gear when returning it?

Goh Mei Ling	Participant refuses to answer this question.
Ali Hakim	Yes, I find it very helpful. If a customer reports visible wear or damage when returning gear, it allows me to address issues quickly before the next rental.
Syed Nasrul	I think it is really useful. A return checklist helps customers ensure all items are intact. This also helps me avoid losses and make the speeds up the checking process.

What features would you like to see in a rental platform to make your job easier?

Goh Mei Ling	I would like a chat function that allows me to ask or answer questions. I also think having clear return instructions that can prevent gear from being mishandled by customers.
Ali Hakim	I'd love to have real-time stock visibility to manage my inventory better, especially during peak seasons.
Syed Nasrul	For me, having options to sort equipment and including user guide would be a big help. It also saves me from answering the same questions over and over.

Do you think bundle gear sets are helpful for users? If so, how should they be grouped or categorized?

Goh Mei Ling	Yes, bundle sets for activity would help users make quicker, easier decisions. Moreover, grouping them by activity type also makes the process less overwhelming.
Ali Hakim	I find equipment bundles based on activity would be very useful as it could help customers to get exactly what they need.
Syed Nasrul	Bundle sets are especially helpful for beginners, as it could help in decision making and simply the selection process.

Do you think a flexible delivery method such as pickup or delivery would help improve your service?

Goh Mei Ling	Absolutely, letting customers choose their delivery location is convenient and saves time. I think this also makes the service more attractive.
Ali Hakim	Yes, I agree. If my customers could select the delivery location based on their preferences, it would reduce a lot of hassle on both ends.
Syed Nasrul	From my point of view, giving customers flexibility enhances the user experience and satisfaction, making them more likely to return.

What role do you think safety information, return instructions, or checklists play in the rental process?

Goh Mei Ling	Safety information is important, especially for water gear. Thus, having clear return instructions helps avoid misuse and damage.
Ali Hakim	Knowing the gear’s condition helps ensure safety. Besides, a proper system that tracks when customers return the gear could also help manage its condition effectively.

Syed Nasrul	Return checklists help reduce missing items and track gear condition. User guides are also important and essential, especially for new users.

Would it be helpful to have a real-time gear availability system?

Goh Mei Ling	Yes, I think it is important, as without real-time availability, there is a risk of double bookings, especially during holidays or weekends.
Ali Hakim	I strongly agree. Real-time stock visibility is a must as it helps avoid spikes in rental demand during peak seasons and public holidays.
Syed Nasrul	The participant has already answer this in Question 8 and would like to skip this question.

Would you consider offering features like rental extension through an app, loyalty programs, or weekend-specific pricing?

Goh Mei Ling	The participants refused to answer this question.
Ali Hakim	Yes, I am interested in all of those. Loyalty programs help encourage repeat customers. I also think rental extensions are useful for customer convenience.
Syed Nasrul	Definitely. I believe rental extensions are very convenient as they save time and help avoid unnecessary communication.